HARD · ADVANCED MATH

Advanced Math

13

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

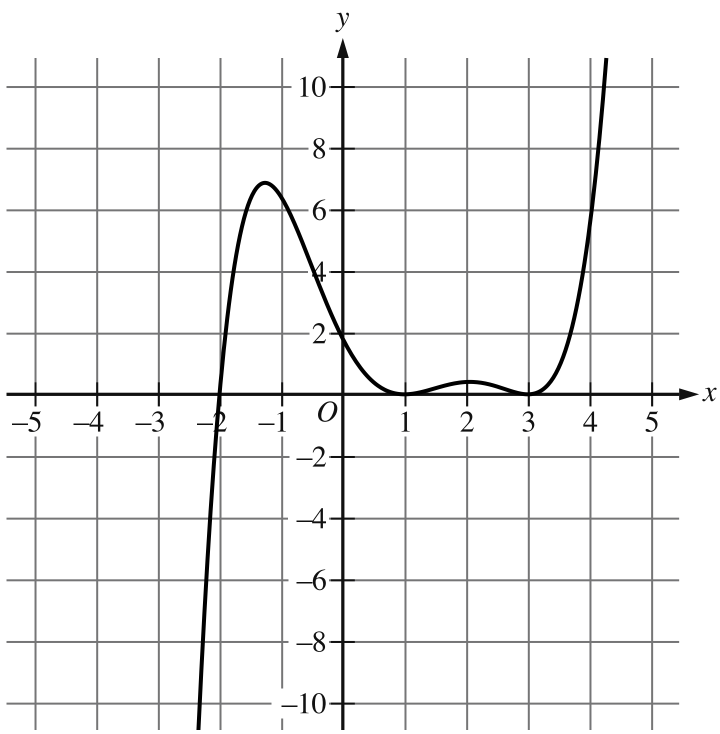
Q1 GRID-IN ADVANCED MATH

In the xy-plane, the graph of $y=3x^2-14x$ intersects the graph of $y=x$ at the points $(0, 0)$ and (a, a). What is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- Moving from left to right, the curve passes from quadrant 3 to quadrant 2 to quadrant 1.
- The curve has 2 relative maximums and 2 relative minimums.
- The curve crosses the x axis at the point (negative 2 comma 0).
- The curve touches the x axis at the following points:
  - (1 comma 0)
  - (3 comma 0)

The graph of $y = gx$ is shown. If $gx + 28 = \frac{1}{10}x - a^2x - b^2x - c$, where a, b, and c are constants, which of the following could be the value of b?

A. 31

B. 3

C. -2

D. -25

Q3 GRID-IN ADVANCED MATH

$$x - 47^2 = 1$$

What is the sum of the solutions to the given equation?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN ADVANCED MATH

$$f x = x^2 - 48x + 2{,}304$$

What is the minimum value of the given function?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5 ADVANCED MATH

$$y = 2x^2 - 21x + 64$$

$$y = 3x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y), in the xy-plane. What is the value of x?

A. -8

B. -6

C. 6

D. 8

Q6

ADVANCED MATH

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy-plane passes through the points $(0, 10)$ and $(-2, \frac{325}{36})$. What is the value of ab?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. 54

D. 60

Q7

ADVANCED MATH

In the xy-plane, the graph of $y = x^2 - 9$ intersects line p at $(1, a)$ and $(5, b)$, where a and b are constants. What is the slope of line p ?

A. 6

B. 2

C. -2

D. -6

$$fx = 9(4)^{x}$$

The function f is defined by the given equation. If $g(x) = f(x+2)$, which of the following equations defines the function g?

A. $gx = 184^{x}$

B. $gx = 1444^{x}$

C. $gx = 188^{x}$

D. $gx = 8116^{x}$

Q9

ADVANCED MATH

$$\frac{20}{p} = \frac{20}{q} - \frac{20}{r} - \frac{20}{s}$$

The given equation relates the positive variables p, q, r, and s. Which of the following is equivalent to q?

A. $p + r + s$

B. $20p + r + s$

C. $\frac{prs}{pr + ps + rs}$

D. $\frac{prs}{20p + 20r + 20s}$

Q10

ADVANCED MATH

$$f x = -6x^2 + 60x - 126$$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the maximum value of the function as a constant or coefficient?

A. $f x = -6x^2 + 42x + 18x - 126$

B. $f x = -6x(x - 7) + 18(x - 7)$

C. $f x = -6x - 5^2 + 24$

D. $f x = -6x - 7x - 3$